

Event Camera Nighttime Performance Degradation

https://gemini.google.com/app/48817205d033e093

User prompt: what is the performance degradation of event cameras at night if only starlight and/or moon light is available?

Response: While event cameras offer massive dynamic range advantages in varying lighting conditions, they suffer severe performance degradation below 1 lux (moonlight) and hit a physical wall in starlight (~0.001 lux).

At these photon-starved levels, the asynchronous pixel architecture cannot accumulate charge fast enough, leading to four primary failures:

- Latency Spikes:** The hallmark advantage of event cameras—microsecond latency—disappears. The photoreceptor circuit's response time is inversely proportional to the photocurrent. A pixel that reacts in <150 μ s in daylight might take tens or hundreds of milliseconds to register a change in moonlight.
- SNR Collapse (Background Noise Storm):** At starlight levels, thermal electrons (dark current) surpass the actual photon-generated current. The sensor begins firing a massive, continuous stream of "false" background events, completely burying the signal of a moving target.
- Contrast Threshold Degradation:** The minimum relative change required to trigger an event effectively rises. A low-contrast object (like a dark drone against a night sky) will fail to trip the comparator threshold, making the target invisible to the event stream.
- Trailing and "Motion Blur":** Because of the slow pixel recovery time at extremely low currents, a moving object leaves a long "comet tail" of events. This mimics the motion blur of standard frame-based cameras, defeating the purpose of high-speed tracking.

Hardware Benchmarks: The 5 Lux Floor

Looking at industry-standard neuromorphic hardware like the Sony Prophesee IMX636, the degradation curve is stark when moving from standard lighting to tactical low-light environments:

Metric	1,000 Lux (Overcast Day)	5 Lux (Deep Twilight)	< 0.1 Lux (Moonlight/Starlight)
Pixel Latency	< 150 μ s	~ 1,000 μ s (1 ms)	Exponentially higher
Background Noise Rate	0.1 Hz per pixel	10 Hz per pixel	Overwhelms signal
Dynamic Range	> 86 dB	> 86 dB	Unreliable

The Cutoff: The absolute minimum low-light cutoff for a sensor like the IMX636 is roughly **0.08 lux** (equivalent to a bright full moon), and performance at this level is not guaranteed by the manufacturer. In starlight (0.001 lux), the sensor is functionally blind.

If you are architecting a multi-modal tracking pipeline for night operations, relying solely on an event camera will fail. You must bridge the gap by either fusing the event stream with Single-Photon Avalanche Diodes (SPADs) or 4D AESA radar, or by introducing active IR illumination (which silicon-based event cameras can detect if the wavelength aligns with their spectral sensitivity).

Sources

- [Canon's new security-focused 'SPAD' camera can capture color](#)
- [IMX636-AAMR-C - Prophesee](#)

User prompt: what are Single-Photon Avalanche Diodes (SPADs)? Is this basically night vision?

Response: No, it is not traditional night vision, though it solves the same low-light problem using completely different physics.

Traditional night vision (like PVS-14 goggles) uses an **analog vacuum tube** (an Image Intensifier or I2 tube). Photons hit a photocathode, release electrons, and those electrons are multiplied across a microchannel plate before smashing into a phosphor screen to create that familiar 2D green glow. It is entirely analog.

A **Single-Photon Avalanche Diode (SPAD)** is a **digital, solid-state semiconductor**. It operates on the principle of extreme amplification at the silicon level, specifically designed to detect the arrival of one single photon with picosecond timing accuracy.

How a SPAD Works (Geiger Mode)

Unlike a standard camera pixel (which accumulates charge over a set exposure time) or an event camera pixel (which fires on a logarithmic change in intensity), a SPAD operates in **Geiger mode**.

- The Bias:** The diode is biased *above* its breakdown voltage. In this highly unstable state, it is essentially a tripwire waiting for the slightest disturbance.
- The Impact:** When a single photon strikes the depletion region, it generates an electron-hole pair.
- The Avalanche:** Because of the high reverse bias, this single carrier is violently accelerated, colliding with other atoms and knocking more electrons loose. This creates a massive, instantaneous "avalanche" of current.
- The Quench:** A quenching circuit immediately drops the voltage below breakdown to stop the avalanche, registers a digital "1", and resets the pixel for the next photon.

Why SPADs Matter for Edge and Tactical Sensors

Because SPADs output a direct, massive digital pulse the exact picosecond a photon arrives, they are fundamentally different from traditional imaging.

Feature	Traditional Night Vision (I2)	SPAD Array
Output Type	Analog visual image (Phosphor screen)	Digital timestamp per photon
Primary Use	Human-in-the-loop 2D visual observation	Machine vision, 3D Depth (dToF), LiDAR
Timing Precision	Continuous (Phosphor decay creates blur)	Picosecond-level (10^{-12} seconds)
Form Factor	Bulky vacuum tubes, high voltage	Solid-state silicon, easily miniaturized

For systems like dismounted infantry defense or counter-UAS tracking, SPADs aren't used to just "see in the dark." They are the core receiving element in **Direct Time-of-Flight (dToF) LiDAR**. Because they react to single photons with picosecond precision, you can fire a microscopic, eye-safe laser pulse and time exactly how long it takes that single photon to bounce off a drone and return.

When you fuse a SPAD array with an event camera or a 4D AESA radar, you combine the ultra-low latency movement tracking of the event camera with the absolute 3D depth and low-light photon-counting ability of the SPAD.

User prompt: are SPADs commercially available to pair with event cameras?

Response: Yes, SPAD arrays are commercially available, though you will generally not find them sold as pre-packaged, plug-and-play hybrids with event sensors. Instead, you purchase the SPAD array as a discrete time-of-flight (ToF) or LiDAR component and handle the sensor fusion yourself on your edge compute hardware.

A few notable commercial options exist in the industrial and scientific markets:

- **Sony's IMX Series:** Sony has heavily commercialized stacked SPAD depth sensors, originally targeting automotive LiDAR and autonomous mobile robots (AMRs). The IMX560, for example, is a 1/2.9-type 100K pixel SPAD sensor that integrates a logic chip and time-to-digital converters (TDCs) directly on the silicon.
- **Pi Imaging SPAD 512:** Billed as the first commercially available high-resolution SPAD camera, offering a 512x512 array specifically for photon-starved environments.
- **Hamamatsu Modules:** They sell bare SPAD modules and arrays designed for scientific and industrial photon counting, offering digital output directly from the module.

Pairing SPADs with Event Cameras at the Edge To actually pair a SPAD with an event camera in a tactical pipeline, both data streams need to be ingested and time-synchronized by a central compute module, such as a Jetson Orin NX 16GB. The event camera stream provides the high-speed regions of interest (ROI), which tells the system exactly which subset of the SPAD array's point cloud to process for depth, saving massive amounts of compute overhead.

When designing this custom payload, the primary challenge is the power budget and thermal management. If you are running an IDS IMX636 camera module, it operates at 5V and pulls a maximum of 400 mA, keeping its peak power consumption strictly under 2.0W. While the event camera is highly power-efficient, active SPAD arrays (especially if paired with the required VCSEL laser emitter for ToF illumination) will introduce a significantly larger power draw. Your edge system will need to manage these concurrent power rails while handling the synchronization between the asynchronous event stream and the high-frequency digital pulses of the SPAD.

Sources

1. [SPAD ToF Depth Sensor 1/2.9" 100KP IMX560 | Products & Solutions](#)
2. [SPAD 512 high-speed photon counting camera - Pi Imaging](#)

User prompt: No I want to dive into how SPADs work in conjunction with event cameras at night. If only starlight is available, essentially event cameras are blind. But if I am looking for that propeller flicker of that drone against a night sky, the event camera is practically useless, so how does an SPAD alleviate this blindness to detect a drone at say 500m out?

Response: To detect a drone's propeller flicker at 500 meters in starlight, the SPAD does not passively "fix" the event camera's blindness by gathering ambient starlight. Instead, the SPAD is deployed as the receiver in an **active LiDAR system**.

By introducing an active illumination source (a laser), you provide the photons that *both* the SPAD and the event camera need to function in zero-lux conditions. Here is how the physics and the sensor fusion work to achieve that 500m detection.

1. The Active Illumination Fix (The Missing Photons)

To get a return at 500m, the SPAD is paired with a Near-Infrared (NIR) laser emitter (typically a VCSEL pulsing at 850nm, 905nm, or 940nm).

Silicon-based event cameras (like the IMX636) are highly sensitive to NIR light. When the LiDAR system pulses the sky to find the drone, it is actively painting the target with NIR photons. The event camera, which was previously starved of photons and drowning in thermal noise, immediately detects these concentrated laser reflections. The active illumination pulls the event camera out of the noise floor, restoring its microsecond contrast detection.

2. The SPAD's Role: Optical Micro-Doppler

At 500 meters, a drone is likely a sub-pixel speck on the event camera's focal plane. While the event camera can now see the illuminated speck moving laterally across the sky, it cannot resolve the physical blades. This is where the SPAD's picosecond timing takes over.

As the laser hits the drone, the rapidly spinning propellers continuously chop the beam. The returning photons bounce back to the SPAD array. Because the SPAD operates in Geiger mode with a Time-to-Digital Converter (TDC), it records the exact arrival time of every single returning photon.

The SPAD data will show a periodic oscillation in the photon return intensity and timing, corresponding exactly to the RPM of the drone's blades. This creates an **optical micro-Doppler signature**. The SPAD confirms that the target is a drone (not a bird or a balloon) purely based on the temporal frequency of the returning photons, without needing spatial resolution.

3. The Fusion Pipeline at the Edge

When you fuse these two data streams on an edge compute module (like a Jetson Orin NX), they perfectly complement each other's weaknesses at long range:

- **The Event Camera (X, Y Tracking):** Freed from starlight blindness by the laser, the event camera handles the high-speed lateral tracking. It provides the tight, low-latency 2D bounding box of the moving target, telling the system exactly *where* to point or process the SPAD array.
- **The SPAD (Z Depth & Classification):** The SPAD array confirms the absolute range (500m) via Direct Time-of-Flight (dToF) and classifies the target as a rotary-wing drone by analyzing the micro-Doppler blade chop in the photon stream.

By actively illuminating the target, you leverage the event camera's asynchronous spatial tracking and the SPAD's temporal depth-and-flicker detection simultaneously.

User prompt: this isn't making a lot of sense. using an event camera to detect that optical "micro doppler" effect of the propeller is one of the strengths of detecting a multi rotor drone against complex backgrounds. But effectively what you are saying at night the SPAD is now taking on that role rather than the event camera, or possibly if I just use an IR or NIR laser and deploy a raster or ellipsoid scan against a wide region, I can effectively get the same result without an SPAD?

Response: You have correctly identified the event camera's primary daylight strength, but your proposal to use a raster-scanning IR laser to replace the SPAD at night breaks down against two physical limits at 500 meters: temporal aliasing and photon return budgets.

Here is why relying purely on a scanned IR laser and an event camera fails for optical micro-Doppler at that range, and what the correct architectural implementation looks like.

The Temporal Aliasing Problem

Optical micro-Doppler requires a *continuous* observation of the target. A drone's propellers typically spin between 100 Hz and 200 Hz. If you use a raster or ellipsoid scan (like a standard mechanical LiDAR) to search a wide area, the laser beam sweeps across the drone and immediately leaves it.

The laser might only illuminate the drone for 1 millisecond per sweep, with sweeps happening at 30 Hz. You cannot extract a 150 Hz continuous wave signature from a target that is only illuminated once every 33 milliseconds. You are sampling the target too slowly, which completely aliases the flicker. The event camera will just see the laser hit and leave, registering a flash, not a propeller chop.

The 500m Photon Budget

Even if you stop scanning and point a floodlight IR laser at the drone to give the event camera a continuous view, you run into the inverse-square law.

A laser beam hitting a drone at 500 meters and reflecting back to your sensor travels 1 kilometer total. The returning light from a small plastic target is incredibly weak.

- An **event camera (like the IMX636)** requires a logarithmic change in *hundreds* or *thousands* of photons to trigger a pixel event.
- A **SPAD array** operating in Geiger mode requires exactly *one* photon to trigger an avalanche.

To make a standard silicon event camera see that reflection at 500m, you would need an actively cooled, high-wattage laser that would consume massive power and violate eye-safety regulations. The SPAD solves the sensitivity problem at the silicon level.

The Correct Architecture: Cueing, Not Scanning

If your goal is to detect a drone at 500 meters at night and confirm it via optical micro-Doppler, you should not use the optical system (event camera or SPAD) to *search* the wide area.

The correct implementation uses a multi-modal handoff:

1. **Search (RF):** Use a 4D AESA radar to continuously scan the wide area. The radar detects the drone, establishes the track, and generates the initial X, Y, Z coordinates.
2. **Cue (Gimbal):** The radar hands those coordinates to a motorized gimbal.
3. **Hold (Continuous Illumination):** The gimbal points a highly focused, *non-scanning* NIR laser directly at the drone, keeping the beam locked on the target.
4. **Confirm (Optical):** With the laser held steady on the target, the SPAD array receives the single-photon reflections and continuously extracts the micro-Doppler blade chop, simultaneously providing dToF range confirmation that an event camera fundamentally lacks.

User prompt: I'm not saying cut out the SPAD. but it is interesting how this could be leveraged to not use a 4D radar in order to preserve non-emission type detection

Response: You are absolutely right to pull the thread on EM silence. If the operational priority is zero-emission (EMCON) to avoid counter-detection, both the 4D radar and the active VCSEL laser can be stripped from the architecture.

This enables a paradigm known as **Passive Single-Photon Imaging** or **Quantum Compressed Sensing (QCS)**.

A SPAD array can detect a drone at range using only ambient starlight through the following mechanics:

1. Ambient Optical Modulation

Instead of bouncing a laser off the target, the system relies on existing ambient light (starlight, atmospheric glow, distant light pollution). As the drone's propellers spin, they mechanically occlude and reveal the body of the drone or the sky behind it, acting as a physical optical modulator. The blades literally chop the ambient starlight.

2. Ultra-Sparse Photon Counting

An event camera fails in these conditions because it requires a minimum threshold of charge accumulation to trigger a pixel event, which is overwhelmed by thermal noise in starlight. A SPAD array detects individual photons with zero readout noise. Even if the SPAD only catches 1 to 10 photons per pixel over several microseconds, those timestamps are recorded with picosecond precision.

3. Frequency-Domain Extraction

At 500 meters in starlight, the drone registers as invisible static in the raw photon stream. Because exact timestamps exist for every photon, the edge system runs a Short-Time Fourier Transform (STFT) on the data rather than attempting spatial image reconstruction.

- **The Noise Floor:** Ambient thermal noise and random starlight photons hit the SPAD randomly, appearing as flat white noise in the frequency domain.
- **The Target Signature:** The photons bouncing off the spinning propeller are modulated at a specific frequency.

The STFT pulls the exact micro-Doppler frequency spike (e.g., 120 Hz) out of the background noise floor. Recent studies demonstrate this passive quantum compressed sensing technique extracting dynamic drone features at up to 10 kilometers without a single active emission.

The Operational Trade-Off

When the active laser pulse is eliminated, the system loses Direct Time-of-Flight (dToF) depth measurement. The passive SPAD provides a precise optical bearing (X, Y) and confirms the target is a rotary-wing drone by its RPM signature, but it cannot calculate the absolute range. To generate a 3D firing solution passively, the architecture requires two geographically separated SPAD arrays triangulating the same micro-Doppler signature.

Sources

1. [Accurate Simulation Pipeline for Passive Single-Photon Imaging](#)
2. [\(PDF\) 10-km passive drone detection using broadband quantum](#)

User prompt: wait... using the same Sony IMX SPAD, this sensor passively at night could detect a quadcopter at up to 10km? How was this achieved?

Response: This was actually proven in a July 2025 breakthrough published in *Light: Science & Applications*, where researchers achieved exactly this: 10-km passive drone detection using a SPAD array and a technique called **broadband quantum compressed sensing (QCS)**.

At 10 kilometers, a drone is less than a single pixel wide, and the ambient light is so weak that the researchers faced a Signal-to-Background Ratio (SBR) of 1/332—meaning for every one useful photon reflecting off the drone, the sensor was bombarded by 332 photons of pure random background noise. Traditional imaging completely fails here because it tries to accumulate enough light to draw a spatial picture (a contour), which is impossible for a point-object buried in that much noise.

Here is how the SPAD array and QCS achieved the 10-km passive detection:

- **Exploiting Quantum Randomness:** Background noise (like starlight, city glow, or thermal dark current) emits photons completely randomly. The drone's spinning blades, however, periodically chop whatever faint ambient light exists. The SPAD array does not wait to form an image; it simply logs the sparse, exact picosecond arrival time of every individual photon that hits the lens.
- **Frequency-Domain Isolation:** Because the background noise is perfectly random, it appears as a flat horizontal line of "white noise" with a constant power spectral density when converted to the frequency domain. The drone's blade chop, however, is rhythmic. By running the sparse photon timestamps through a Fourier Transform, the specific dynamic frequency of the propeller spikes prominently out of that flat noise floor.

- **Extreme Data Compression:** By treating the SPAD as a time-domain frequency sensor rather than a spatial camera, the system achieved a massive data compression ratio of 1.95×10^{-6} . This allowed the researchers to detect dynamic frequencies up to 2.05 GHz through incredibly sparse photon detection without requiring massive computational overhead.

In short, they stopped trying to "see" the drone spatially and used the SPAD to listen to the quantum rhythm of the photons. As long as the propellers are chopping ambient light, the SPAD can extract that frequency from the noise passively.

Sources

1. [\(PDF\) 10-km passive drone detection using broadband quantum](#)
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